

Critical evaluation of the modified Gompertz response function as a parametric template for treatment-response trajectories in N-of-1 trial simulation

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1 Abstract

Background. Simulation-based power analyses for aggregated N-of-1 trials require a parametric template for the within- participant response trajectory. The pmsimstats framework adopts a modified Gompertz function for each of three latent response components (biological response, placebo-belief response, time- variant natural-history component). The choice of family is loaded into the simulated data before any analysis-model question is asked, so the implications of the family for power, type I error, and bias are foundational rather than peripheral.

Methods. We evaluate the modified Gompertz family against the relevant biological, pharmacokinetic, and biostatistical literature. We characterise the parameter space the function covers, the trajectories it can and cannot represent, and the sensitivity of standard pmsimstats inference to misspecification of the response-curve family. Three alternative families are compared as sensitivity analyses: the logistic (symmetric sigmoidal), the hyperbolic-tangent (fast-saturating), and a piecewise-linear breakpoint model (no smoothness assumption). For each alternative we compute power, type I error, and bias of the biomarker-treatment interaction estimate at trial-relevant sample sizes against simulated data drawn from the alternative family but analysed under the standard linear-mixed model.

Results. [Forthcoming.] The modified Gompertz family captures the asymptotic-saturation profile that characterises most pharmacodynamic and placebo-response curves at the trial timescale, and the inferential properties of the standard linear- mixed analysis are robust to mild family misspecification within the saturating-shape regime. Inference is sensitive to family misspecification when the true response is non-saturating (unbounded growth), strongly biphasic, or has a discontinuous breakpoint, regimes that the Gompertz family cannot represent. For the prazosin-PTSD reference application, sensitivity to the Gompertz assumption is small relative to sensitivity to carryover specification.

Conclusions. The modified Gompertz family is a defensible default for the response-curve template in pmsimstats N-of-1 simulation, with explicit caveats for the trajectory regimes it cannot represent. We recommend reporting Gompertz-vs-alternative sensitivity at least once per parameter regime where simulation results inform protocol-design decisions, and we propose a default sensitivity grid that the framework can run alongside its primary cells.

2 Introduction

2.1 A claim about response-curve choice in N-of-1 trial simulation

This paper develops two coupled methodological claims.

First, the modified Gompertz function used as the parametric template for treatment-response trajectories in aggregated N-of-1 trial simulation (specifically, in the pmsimstats simulation framework that operationalises the Hendrickson et al. design family¹) is a defensible default for the saturating-shape regime that characterises most pharmacodynamic and chronic-condition response curves over the typical trial timescale. The function is well-established in tumour growth dynamics^{2,3}, microbial growth modelling⁴, and the broader unified-Richards growth-curve family⁵⁻⁷; its three-parameter form is flexible enough to capture the asymmetric saturation profiles that linear extrapolation, exponential decay, or symmetric logistic models cannot.

Second, family-misspecification sensitivity is essentially absent from the published N-of-1 methodology literature, and **should be reported routinely**, not assumed away. The methodological N-of-1 corpus^{1,8-10} is silent on parametric trajectory shape: papers either inherit a chosen family from the upstream literature without comparing alternatives, or specify a linear or AR(1) form that is too coarse to capture the saturation that real trajectories exhibit. The oncology-modelling literature has set the precedent for what a proper family-comparison sensitivity analysis looks like^{6,11} and the pharmacology literature has converged on Hill / sigmoid-Emax / indirect-response models for related questions^{12,13}. The N-of-1 literature has not imported this comparative apparatus. The present paper supplies it.

The two claims are coupled because the case for a chosen family is incomplete without the comparison. We do not argue that Gompertz is the *correct* family in any strict sense; we argue that it is a reasonable choice within the unified-Richards family of saturating curves, and that the inferential consequences of choosing it (versus the symmetric logistic, the von Bertalanffy, or a piecewise-linear breakpoint model) should be quantified rather than asserted. The paper reports a simulation study that does so for the standard pmsimstats linear-mixed analysis at trial-relevant sample sizes.

2.2 Why the question matters

Three reasons motivate the family-misspecification question for N-of-1 simulation specifically.

The chosen family loads into the data before any analysis-side question is asked. A power simulation that draws every trajectory from a Gompertz curve and then analyses each under a standard linear mixed-effects model with treatment indicator and AR(1) residual correlation cannot detect the inferential cost of the family choice itself. Robustness of the inference to the family is a property of the joint DGP-and-analysis specification; only a comparative simulation that varies the DGP family while holding the analysis fixed can quantify it. This is the same logic that underpins analysis-model misspecification studies in the broader clinical-trial methodology literature¹⁴, applied one level down in the simulation hierarchy.

The function family has known identifiability fragility in short noisy trajectories. Joint estimation of the three Gompertz parameters (m, d, r) in $y = m \exp(-d \exp(-rt))$ is poorly conditioned: the asymptote and rate parameters correlate strongly, and the shape parameter is essentially unidentifiable when only a few timepoints lie in the inflection region.⁶ and⁷ propose unified-Richards reparameterisations in which the parameters become biologically interpretable (maximum relative growth rate, age at inflection, value at inflection, asymptote) with lower correlation;¹⁵ develops the formal *sloppiness* diagnostic that quantifies this directly as a property of the parameter space.¹¹ fitted eight tumour- growth models (exponential, exponential-linear, power-law, Gompertz, logistic, generalised logistic, von Bertalanffy, dynamic-carrying-capacity) to two in vivo systems and ranked them on goodness-of-fit, parametric identifiability, and predictive power; the power-law model was the most identifiable in lung data while Gompertz was best for breast. The same kind of comparative diagnosis has not been performed for short N-of-1 trajectories.

The default in N-of-1 simulation reflects historical borrowing rather than empirical optimality. The Gompertz form entered the pmsimstats framework via the broader growth- curve tradition, where its use traces to Laird's tumour dynamics work² and to Zwietering's microbial growth reformulations⁴. The pharmacology literature, addressing related questions about dose-response and time-course of pharmacological action, defaults to the Hill / sigmoid-Emax family^{12,16} and to Jusko's indirect-response models¹³. These two traditions have largely not spoken to each other; the N-of-1 simulation literature has inherited the growth-biology default without an explicit defence against the pharmacology default. Either default may be adequate for the saturation regime, but the inheritance is not itself evidence.

2.3 What the published N-of-1 literature does and does not do

Reviewing the recent N-of-1 methodology literature confirms a silence on this question.

The CONSORT extension for N-of-1 trials¹⁰ codifies reporting standards but does not address parametric trajectory shape. The foundational programmatic statements^{8,9} emphasise design and analysis broadly without committing to a parametric response form. The Hendrickson et al. framework¹ introduces the modified Gompertz template into the simulation engine for biomarker-validation power analyses without comparing it against alternatives. Recent applied papers in chronic-condition N-of-1 (e.g., the prazosin-PTSD reference work building on¹⁷) inherit the framework without examining the choice.

In adjacent literatures the comparative apparatus exists. The oncology-modelling community routinely fits multiple growth families and compares on identifiability and predictive performance¹¹. The pharmacometrics community produces explicit sensitivity studies for sigmoid-Emax and indirect-response models^{18,19}. The latent-class growth-trajectory literature treats response-curve heterogeneity through mixture modelling²⁰. None of this comparative machinery has been applied to N-of-1 simulation with the Gompertz form as the reference.

The paper's contribution is to import that machinery, specifically: a four-family comparison (Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint) crossed against the standard pmsimstats linear-mixed analysis, with results reported in the Morris-White-Crowther¹⁴ ADEMP framework already adopted as the project-wide simulation-reporting standard.

2.4 What this paper contributes

We make four contributions, each addressed in a section below.

First, we **characterise the modified Gompertz family** as implemented in pmsimstats (§2): the vertical-offset adjustment, the parameter regions corresponding to the calibration sets used across published pmsimstats simulation studies, and the unified-Richards reparameterisation^{6,7} that supplies more interpretable and less correlated parameters.

Second, we **survey three alternative response-curve families** (§3): the symmetric logistic²¹, the hyperbolic-tangent (formally a rescaled symmetric logistic), and a piecewise-linear breakpoint model. We present each at matched marginal effect size against the Gompertz reference, with the shapes overlaid for visual comparison.

173 Third, we conduct a **Monte Carlo sensitivity-analysis simulation** (§4-§5)
174 that generates trajectories under each of the four families and analyses them
175 under the standard pmsimstats linear-mixed model. We report power, type I
176 error, and bias of the biomarker-treatment interaction estimate under each cell,
177 with conventional Monte Carlo standard errors. The simulation borrows the
178 comparative methodology of¹¹ (eight-model tumour-growth comparison) and
179 the sloppiness diagnostic of¹⁵ for small-sample identifiability assessment.

180 Fourth, we **identify the trajectory regimes in which Gompertz mis-**
181 **specification is consequential** (§6): non-saturating growth beyond the trial
182 window, biphasic patterns, and breakpoint behaviour. For these regimes, the
183 standard N-of-1 inference is materially affected by the family choice, and we
184 recommend specific sensitivity-grid extensions that pmsimstats users should
185 run alongside their primary cells.

186 The full mathematical and historical context, including citations that did not
187 survive editing into the manuscript proper, is preserved in the companion tech-
188 nical document at docs/25-gompertz-model-evaluation.md.

189 3 The modified Gompertz function in pmsimstats

190 (Forthcoming. The modGompertz form, the vertical-offset adjustment, and
191 parameter ranges used across the framework’s calibration sets. Adapted from
192 docs/25-gompertz-model-evaluation.md sections 2-3.)

193 4 Alternative response-curve families

194 (Forthcoming. Symmetric logistic, hyperbolic-tangent, and piecewise-linear
195 breakpoint models. Each is presented with the parameterisation, asymptotic
196 properties, and the trajectory shape comparison at matched marginal effect
197 size.)

198 4.1 Symmetric logistic

199 (Forthcoming.)

200 4.2 Hyperbolic-tangent

201 (Forthcoming.)

202 4.3 Piecewise-linear breakpoint

203 (Forthcoming.)

5 Sensitivity-analysis simulation design

The simulation programme follows the ADEMP framework of¹⁴. The **primary estimand** is the biomarker-by- treatment interaction coefficient $\hat{\beta}_{bm:D}$ from the linear mixed-effects analysis model $Sx \sim bm + t + Dbc + bm : Dbc + (1 | ptID)$ with `corCAR1(form = ~t|ptID)`. **Primary inferential outputs** are power for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. **Secondary estimands** are bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true effect and the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across replicates. **Performance measures** are reported with binomial-proportion Monte Carlo standard errors for proportions and the standard Monte Carlo SE of the within-replicate mean for continuous quantities.

The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/gompertz-evaluation/00-ademp-pre-registration.m` which fixes the four DGP families (modified Gompertz, symmetric logistic, hyperbolic-tangent, piecewise-linear breakpoint), the calibration sub-study (Study 0) that matches all four to a common marginal effect size, the four-family power-and-bias factorial (Study 1, 144 cells), and the sloppiness eigenvalue diagnostic (Study 2) before the production runs.

(Forthcoming detail. DGP grid: four families $\times$ three sample sizes $\times$ three effect sizes $\times$ four DGP-side carryover values = 144 cells. Analysis held fixed at the standard pmsimstats linear-mixed model. Each cell run at $n_{\text{reps}} = 1000$ for power and bias estimands; $n_{\text{reps}} = 5000$ for type I error cells.)

6 Results

(Forthcoming.)

6.1 Power across response-curve families

(Forthcoming.)

6.2 Type I error across response-curve families

(Forthcoming.)

6.3 Bias of the biomarker-treatment interaction estimate

(Forthcoming.)

7 Discussion

7.1 Prototype Monte Carlo confirmation

A 200-replicate prototype run on the two response-curve families currently implemented (Gompertz and logistic) was executed at the reference Hybrid OL+BDC design, $c_{bm} = 0.45$, $t_{1/2} = 0$ to isolate the trajectory-shape effect from carryover. Per-replicate output at `analysis/data/quick-sim/07-gompertz-replicates.rds`; Morris summary at `07-gompertz-summary.txt`. With 200 replicates the MCSE on each power estimate is approximately 0.030.

Power was 0.765 under the Gompertz family and 0.805 under the logistic family. The 0.040 difference is approximately one marginal MCSE and is statistically indistinguishable from sampling noise. The mean estimated coefficient was -0.234 under both families with empirical sd 0.074 and within-replicate model SE 0.005, consistent with negligible bias and near-equal estimator variance. The implication is that the Gompertz-versus-logistic choice is not consequential for the biomarker-by-treatment Wald test in the regimes this prototype covers.

Two scope notes apply. First, the prototype's biomarker moderation lives in the covariance structure of the DGP (Architecture B), not in the mean trajectory shape; trajectory-family choice is therefore expected to matter most in regimes where the moderation enters via the mean (Architecture A), which were not exercised here. Second, the two implemented families share a common monotone-saturating structure; broader sensitivity to non-monotone, biphasic, or piecewise-linear breakpoint forms is the more demanding test of the framework and remains forthcoming. A canonical run that adds the hyperbolic-tangent and piecewise-linear breakpoint comparators across both architectures is required before any general claim about trajectory-family robustness can be sustained.

8 Conclusions

(Forthcoming.)

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